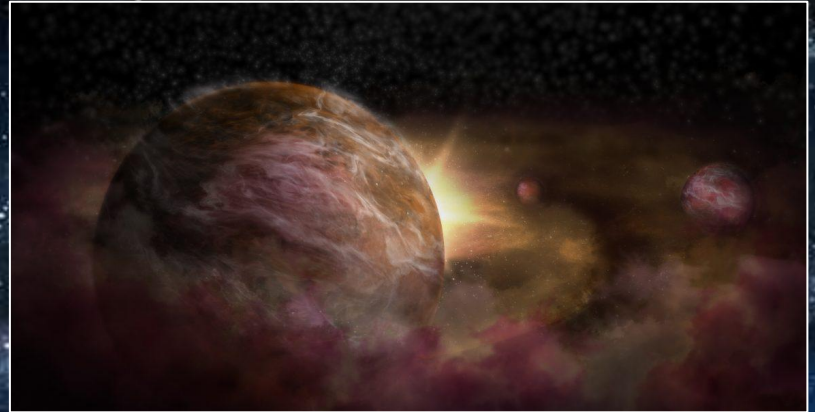


Discovering an Exoplanet

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Research Goal

- The goal of our project is to use some of what we have learned about coding and apply it to real world research
- Our real world research goal is to prove the existence of an exoplanet
- The code we worked on will show how we can use what we learned these last six weeks to find an exoplanet



Background Research

Can we confirm a planet exists using only a star's brightness over time?

11th century — Avicenna records a dark spot crossing the Sun, likely Venus in transit

1952 — Otto Struve formally proposes transits as a planet-detection method

The 50-year gap: a Jupiter blocks $\sim 1\%$ of a star's light, an Earth $\sim 0.008\%$ — too small for ground-based telescopes

2000 — first exoplanet confirmed by transit

2009 — Kepler launches, monitors $\sim 150,000$ stars at once

$\sim 75\%$ of exoplanet discoveries since 2016 come from transits (Deeg & Alonso, 2018)

Why it matters: transits are the only method that gives us planetary radius.
Radius + mass $\rightarrow$ density $\rightarrow$ rock, ice, or gas.

Background Research

What Answering This Requires

Precision — detect a $\sim 1\%$ brightness dip. Requires space-based photometry

Detrending — stars rotate and have spots; the baseline drifts more than the transit is deep

Periodicity — one dip is noise. A repeating dip is an orbit. Needs a period-search algorithm

False-positive checks — eclipsing binaries mimic transits. Check dip shape and odd vs. even depths

Steps

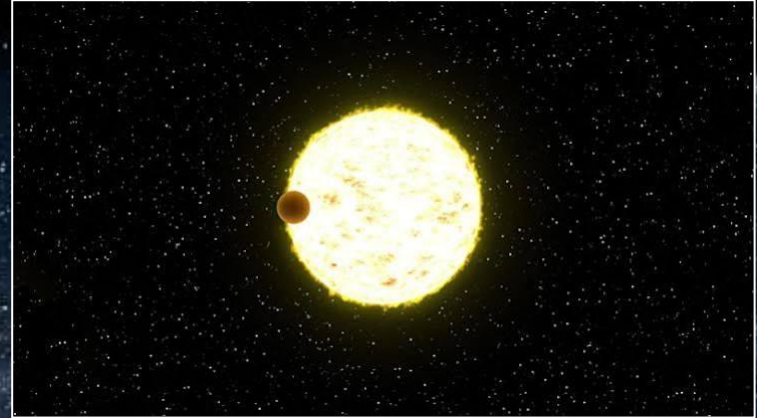
The steps we took to prove the existence of an exoplanet are:

- Choosing a method
- Finding the Data
- Cleaning and adjusting the data
- Graphing and analyzing the data

Methods

The different methods to find an exoplanet are:

- The transit method
- Radial velocity (doppler)
- Direct Imaging
- Microlensing



We choose the transit method because it is the most reliable method and the most common method for finding an exoplanet. It also doesn't have many limitations as opposed to other methods.

Our Analysis

1. Data Acquisition & Cleaning

- **lightcurve** - downloads Kepler photometry, removes outliers/NaNs, normalizes flux

```
lc = search_result[0].download()

lc_clean= lc.remove_nans().remove_outliers(sigma=5).normalize()

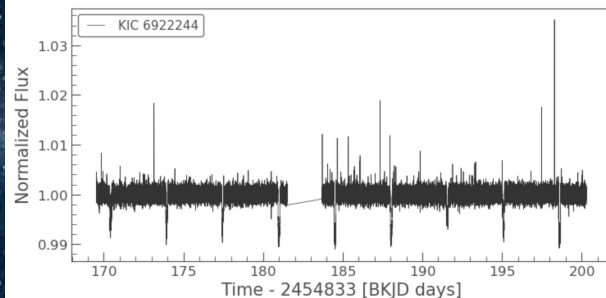
lc_clean.plot()
plt.show()
```

2. Detrending

- **.flatten()** - removes long-term stellar variability, keeps the short transit dips

```
lc_flat = lc_clean.flatten(window_length=401)

lc_flat.plot()
plt.show()
```



Our Analysis

- **Transit Detection**

- **Box Least Squares (BLS) periodogram** - searches for the periodic dip signature; returns orbital period, duration, depth

```
lc_folded = lc_flat.fold(period=peak, epoch_time=periodogram.transit_time_at_max_power)
lc_folded.scatter()
```

- **Radius Calculation using Transit depth**

- planet radius, using $R_p = R_* \sqrt{\delta}$

- **Transit Timing & Companion Search**

- **scipy.curve_fit** + N-body simulation (**solve_ivp**) — fits each transit's exact timing, then simulates hypothetical companion planets to place a mass upper limit

- **Physical Characterization**

- Distance from star; habitable zone, and equilibrium temperature

Choosing and Cleaning Data

- The star we be looked at is Kepler 8

- We chose to use data from the kepler 8 star because it's a good, clearcut example of how to transit method works. It is also located in our galaxy. So it will be relatively convenient to study.

- Cleaning data is an important part of our project because it allows us to filter out unnecessary information that could potentially lead to incorrect data

Transit Method

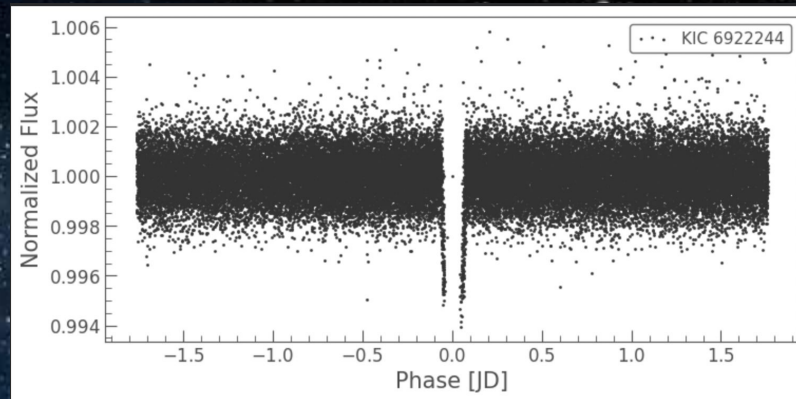
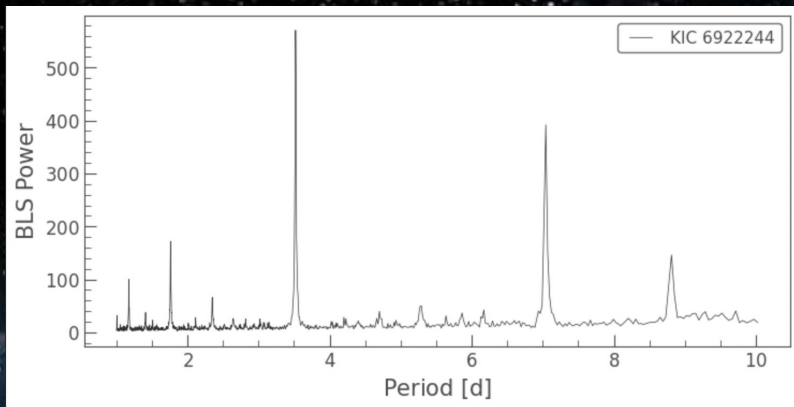
What is the Transit method?

- A planet passing in front of its star blocks a little bit of light
- That creates a small, repeating dip in brightness
- Dip depth tells us the planet's size
- Time between dips tells us the orbital period
- Small shifts in dip timing can hint at another hidden planet

What we're looking for in our code:

- A repeating dip we can pull out of the noise
- A reliable period
- Precise enough timing to check if the dips are on schedule

Transit Method





Proving the data reveals an exoplanet

- Strong, unambiguous periodic signal: the BLS periodogram peak at 3.52 days reaches a power of ~ 570 , far above the surrounding noise floor (~ 10 -50) — not a chance fluctuation
- Clean, repeating transit shape: folding and binning the light curve at that period produces a clear box-shaped dip, flux dropping from 1.000 to ~ 0.9955 ($\sim 0.45\%$ depth) and recovering — consistent across the full dataset, not a one-off event
- Independent confirmation: the measured 3.52-day period matches the published orbital period of the real, already-confirmed exoplanet Kepler-8b (Jenkins et al., 2010) — this pipeline recovered a known result from scratch using nothing but the raw NASA brightness data

Calculating Radius of Planet

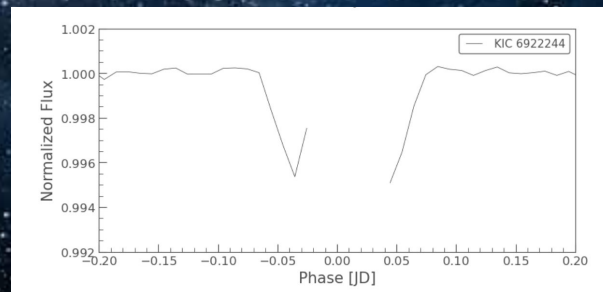
Formula

```
#Radius Calculation
# fractional flux difference = (radius of planet / radius of star)**2
#(F_normal - F_dip) / F_normal = (Rp / Rs)**2
```

#calculating the dipped flux

```
dip_mask = (lc_folded.time >= -0.04) & (lc_folded.time <= -0.03) | (lc_folded.time >= 0.04) & (lc_folded.time <= 0.05)
dip_flux = lc_folded.flux[dip_mask]
F_dip = np.percentile(dip_flux, 1) #taking the bottom 1 percent for a more accurate flux_dip value
```

- Calculated radius: 0.115 solar radii
- Actual radius (Kepler 8b): 0.146 solar radii
- Error = 21.0%
 - Mainly due to limb darkening



Ruling Out a Hidden Planet (Pt. 1)

- The transit method can't directly calculate mass
- But an N-body simulation can help us rule out possible hidden planets and better understand the system (potentially to calculate the mass)
- Setting up for the simulation:
 - Slicing the data into individual windows of each cycle
 - Finding center time of each dip

```
epochs: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
times: [169.53112200414762, 173.0520490143007, 176.5729760244538, 180.09390303460688, 183.61483004475997, 187.13575705491309, 190.65668406506617, 194.17761107521926, 197.69853808537235, 201.21946509552544] days
```

```
#creating time windows for each dip
windows = []
mask = 0
for j in times:
    left_bound = j - 0.12 #a window range that leaves room for potential time variations
    right_bound = j + 0.12
    mask = (lc_flat.time.value >= left_bound) & (lc_flat.time.value <= right_bound)
    windows.append(lc_flat[mask])
```

	orbit number	middle time	uncertainty
0	0	169.462195	inf
1	1	173.073854	0.002076
2	2	176.555305	0.002026
3	3	180.091329	0.002032
4	4	183.614830	inf
5	5	187.137044	0.002037
6	6	190.659990	0.002041
7	7	194.179321	0.002031
8	8	197.687067	0.002033

Ruling Out a Hidden Planet (Pt. 2)

The Assumption

- Our radius calculation assumes Kepler-8b is the only planet in the system

Building the Test

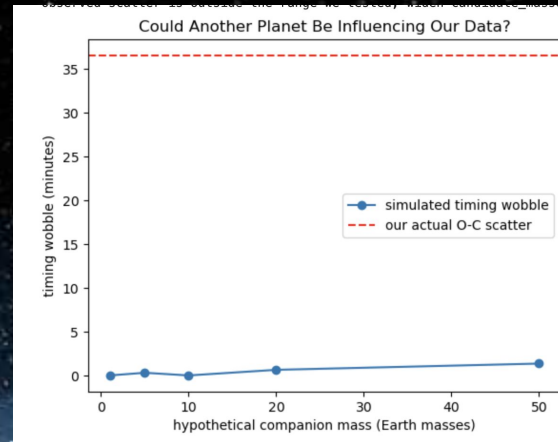
- Built our own gravity simulator using scipy
- Validated it by confirming it reproduces Kepler-8b's real known orbital period

Testing for a Companion

- Simulated how a hidden second planet would affect our transit timing
- Compared the simulated timing wobble to our actual data's timing scatter

Result

- Our timing scatter (~36 minutes) is larger than what a modest companion would cause, meaning it's likely dominated by measurement noise rather than a real planetary signal
- We don't currently have precise enough timing to rule out or confirm a hidden companion, but our earlier radius calculation still holds since we found no clear periodic timing pattern pointing to one



Habitability

Distance From its Star (semi-major axis)

$$P^2 = \frac{4\pi^2}{k^2(M_{\text{tot}})}a^3$$

- We can use Kepler's 3rd law, plugging in the orbital period we already found from the periodogram plus the star's known mass.
- Solving for a , we determined the length of the semi-major axis to be 0.0483 AU
- This is about 8x closer than Mercury is to the sun

Equilibrium Temperature:

$$T_p = T_{\odot}(1 - a)^{1/4}\sqrt{\frac{R_{\odot}}{2D}}$$

- We can assume a very low albedo based on the planet's calculated size and orbital radius
- After simplifying the equation we get an equilibrium temperature of 1661.7 K.
- Far too hot to allow for liquid water, obviously **not habitable**

Habitability (Pt. 2)

Stellar Luminosity:

$$L = 4\pi R^2 \sigma T^4$$

- Luminosity scales with the star's surface area and scales extremely sensitive to its temperature
- We just care about the ratio between the stellar luminosity of Kepler-8 and our sun which we determined to be about 2.96

```
L_star = 4 * np.pi * R_star_q**2 * const.sigma_sb * T_star_q**4
L_ratio = (L_star / const.L_sun).decompose().value
print("L_ratio:", L_ratio)

d_inner = np.sqrt(L_ratio / 1.1) # AU
d_outer = np.sqrt(L_ratio / 0.53) # AU

print(f"Habitable zone: {d_inner:.3f} - {d_outer:.3f} AU")
print(f"Planet's distance: {a_AU.value:.4f} AU")

print("The planet lies well OUTSIDE the habitable zone.")

# insolation flux relative to Earth
S_ratio = L_ratio / (a_AU.value)**2
print(f"\nInsolation flux received: {S_ratio:.1f}x what Earth receives from the Sun")
```

Habitable Zone:

$$d_{\text{inner}} = \sqrt{\frac{L_{\text{ratio}}}{1.1}} \text{ AU}$$

$$d_{\text{outer}} = \sqrt{\frac{L_{\text{ratio}}}{0.53}} \text{ AU}$$

- We scale the Sun's habitable zone boundaries (roughly 0.95–1.4 AU) by how much brighter or dimmer Kepler-8 is.
- We computed a habitable zone of 1.642 - 2.365 AU
- Our planet's semi-major axis (0.048 AU) is well outside that zone

```
L_ratio: 2.9643993650510216
Habitable zone: 1.642 - 2.365 AU
Planet's distance: 0.0483 AU
The planet lies well OUTSIDE the habitable zone.

Insolation flux received: 1270.4x what Earth receives from the Sun
```

Conclusion/ Why this Matters

- Coding is very prevalent in modern science and research

- With advancements in instrumentation, research on exoplanets continues to be on the rise. Coding is a very integral part of it.

- Overall, through what we learned in the last six weeks, we were able to use our skills for real world applications and research; our code has sufficiently proven the existence of an exoplanet



Links/References

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THANK YOU